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RV COLLEGE OF ENGINEERING®
(Autonomous Institution affiliated to VTU)
First Semester B. E.

DEPARTMENT OF MATHEMATICS
FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS
(BRANCHES: ME, AS, IEM, CH)
MODEL QUESTION PAPER - I

Maximum Marks: 100

Time: 03 Hours

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 & 4, 5 & 6, 7 & 8 and 9 & 10.

PART-A

1	1.1	If A is a matrix of order 3×5 then rank of A cannot exceed _____.	1
	1.2	The Eigenvector corresponding to the Eigenvalue $\lambda = 6$ of the matrix $A = \begin{bmatrix} 8 & -4 \\ 2 & 2 \end{bmatrix}$ is _____.	1
	1.3	If two characteristic roots of the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ are 3 and 6, then the third characteristic root is _____.	2
	1.4	Curvature of a straight line is _____	1
	1.5	If two curves intersect orthogonally in cartesian form, then the angle between the same two curves in polar form is _____.	1
	1.6	The angle between the radius vector and tangent for the equiangular spiral $r = ae^{\theta \cot \propto}$ is _____.	2
	1.7	If $z = (\cos x)^y$, then $\frac{\partial z}{\partial y} =$ _____.	2
	1.8	Given that $z = 2xy - 3x^2y$ and x increases at the rate of 2cm/sec. Find the rate at which y changes at the instant when $x = 3$cm and $y = 1$cm in order that the z shall be neither increasing nor decreasing.	2
	1.9	Evaluate $\int_0^2 \int_1^3 \int_1^2 xy^2z \, dx \, dy \, dz$	2
	1.10	Transform the integral $\int_0^a \int_0^{\sqrt{1-x^2}} xydx \, dy$ into polar form.	2
	1.11	If $D = \frac{d}{dt}$ then $\frac{1}{D^2+1}\cos(t) =$ _____.	2
	1.12	Reduce $x^2y'' - 2y = \log(x)$ to differential equation with constant coefficients.	2

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PART-B

2	a	Compute the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$.	5
	b	The currents i_1, i_2, i_3 in the paths of an electrical network follow the linear equations $i_1 - i_2 + i_3 = 0, 3i_1 + 2i_2 = 7, 2i_2 + 4i_3 = 8$. Determine i_1, i_2, i_3 using Gauss-Jordan elimination method.	5
	c	Determine the largest eigenvalue and the corresponding eigenvector of the matrix $A = \begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix}$. Choose the initial vector as $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Perform 5 iterations.	6
3	a	Find the angle of intersection of the pair of curves $r = a\sin\theta$ and $r = \frac{a}{\sin\theta}$.	8
	b	Find the radius of curvature of the tractrix $x = a[\cos t + \log\tan(t/2)]$, $y = a\sin t$.	8
		OR	
4	a	Show that the circle of curvature at the origin for the curve $y = mx + \frac{x^2}{a}$ is $x^2 + y^2 = a(1+m^2)(y - mx)$.	8
	b	Obtain the Maclaurin's series expansion of $e^{\tan^{-1}x}$ upto the term containing x^5.	8
5	a	If $v = (x^2 + y^2 + z^2)^{m/2}$, then find the value of $m(m \neq 0)$ which will make $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$.	8
	b	Show that by Lagrange's method of undetermined multipliers the rectangular solid of maximum volume that can be inscribed in a sphere is a cube.	8
		OR	
6	a	If $z = x^2 \tan^{-1}\left(\frac{y}{x}\right) - y^2 \tan^{-1}\left(\frac{x}{y}\right)$, verify that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$.	8
	b	If $x = uv$, $y = \frac{u}{v}$, then prove that $JJ' = 1$	8
7	a	By changing the order of integration, evaluate $\int_0^a \int_{\frac{y^2}{a}}^{y} \frac{y}{(a-x)\sqrt{ax - y^2}}\, dx\, dy$	8
	b	Evaluate $\iiint_R (x^2 + y^2 + z^2)\, dx\, dy\, dz$ where R denotes the region bounded by $x = 0, y = 0, z = 0$ and $x + y + z = a\ (a > 0)$.	8
		OR	
8	a	Find the volume of a solid bounded by the spherical surface $x^2 + y^2 + z^2 = 4a^2$ and the cylinder $x^2 + y^2 - 2ay = 0$.	8
	b	Determine the centre of gravity of the triangular lamina bounded by the coordinate axis and the line $x + y = 1$.	8

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9	a	The charge $q = q(t)$ on the capacitor in the LRC series electrical circuit is governed by $L\,q'' + Rq' + \frac{1}{C}q = E(t)$, determine $q = q(t)$ if $L = \frac{5}{3}\,H$, $R = 10\Omega$, $C = \frac{1}{30}\,\mathrm{F}$, $E(t) = 300V, q(0) = 0C$, $i(0) = 0A$.	8
	b	Solve $4y'' + 36y = cosec\,(3x)$ using method of variation of parameters.	8
		OR	
10	a	Solve $x^2y'' - xy' + y = 2\log(x)$.	8
	b	The displacement of a mass/spring system is governed by $\frac{d^2x}{dt^2} + \lambda\frac{dx}{dt} + \omega^2x = 0$, where λ and ω are force constants, determine $x = x(t)$, under the conditions $x(0) = 1 = x'(0)$, and $\lambda = 5, \omega = 4$.	8

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